

## Trainings ▾

### Preparatory Steps

**Note :** The following procedure is **only** to be used for diesel engines.

- Remove the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/18/18-003-011-tr.html).

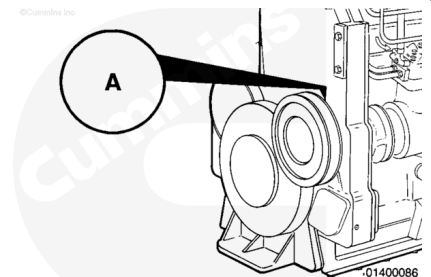


### Adjust

The barring device used on later engines with the one-piece front cover turns approximately two revolutions before the engine begins to turn. The device will **not** turn the engine opposite the direction of normal rotation.

The barring device used on older engines with the two-piece front cover will turn the engine opposite the direction of normal rotation.

Remove the clip and push the shaft in and turn the barring device **counterclockwise** until the "A" mark on the pulley is aligned with the mark that is cast into the boss for the accessory drive seal on the front gear cover.



Determine the cylinder in position for valve set.

The valves will be adjusted on the cylinder that has all of the valves closed.

Use the table to determine the cylinders for valve position.

| If Valve Set Mark Is: | Check Valve Position On: |
|-----------------------|--------------------------|
| A                     | 1,6                      |
| B                     | 2,5                      |
| C                     | 3,4                      |

If the rocker lever assemblies have been removed, use this step to determine the cylinder to be set.

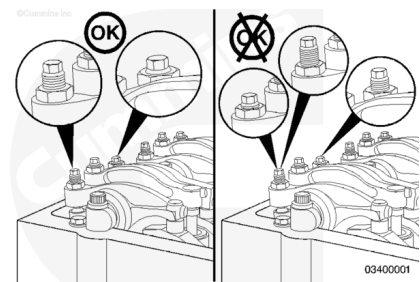
Lubricate the adjusting screw threads with clean engine oil prior to making valve and injector adjustments.

All adjusting screws **must** be loose on all cylinders, and the push rod **must** remain in alignment.

Perform this step on both cylinders to be checked.

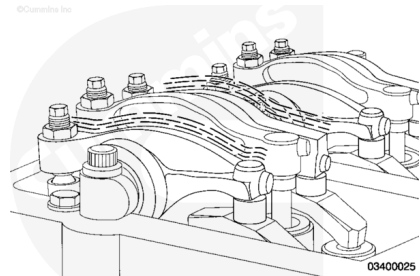
Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the lock nuts until they touch the rocker levers.

The push rods will be the same height above the top of the rocker lever housing on the cylinder ready for valve adjustment.



**⚠ CAUTION ⚠**

**There are two different procedures for adjusting the valves and injectors on K19 engines. Engines with fixed time are adjusted on the inner base circle of the camshaft using a dial indicator. Engines with STC (step timing control) or hydraulic variable timing are adjusted on the outer base circle using a torque wrench. Make sure to use the correct procedure for the engine being serviced, or all of the push rods can be bent.**



If the rocker levers have **not** been removed, wiggle the valve rocker levers on the two cylinders in question.

Set the valves on the cylinder where both rocker levers feel loose.

Use the chart to determine the valve and injector that is ready to adjust.

Adjustment can begin on any valve set mark.

In the example, assume the A mark is aligned and the push rod heights indicate the valves on cylinder number 6 are closed, and one valve on cylinder 1 is fully open.

The chart shows the valves on cylinder number 2 and injector on cylinder number 3 are ready to adjust.

K19 OBC

|     | Valves Closed On | Set |     |
|-----|------------------|-----|-----|
|     |                  | V   | I   |
| A   | 1                | 5   | 4   |
| B   | 5                | 3   | 1   |
| C   | 3                | 6   | 5   |
| (A) | (6)              | (2) | (3) |
| B   | 2                | 4   | 6   |
| C   | 4                | 1   | 2   |

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After the adjustment, bar the engine to the B set mark. Adjust the valves on cylinder number 4 and injector on cylinder number 6.

## Injectors

Use a dial-type torque wrench, Part Number 3164795, or equivalent, to tighten the injector rocker lever adjusting screw. If the screw chatters during setting, repair the adjusting screw and rocker lever as required.

Position the torque wrench so the dial can be viewed in a direct line. This will enable the dial to be read accurately.

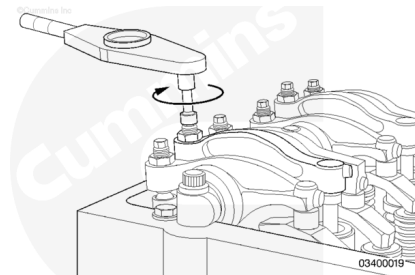
Make sure the parts are in alignment and squeeze the oil out of the valve and injector train, while tightening the adjusting screw. This is an initial preload to the valve train and the injector.

Tighten the adjusting screw.

The torque wrench **must** be calibrated and have a resolution of 0.28 N•m [2.5 in-lb]. and have a range of 17 to 23 N•m [150 to 200 in-lb].

### Torque Value:

1. 11 N•m [ 100 in-lb ]
2. Loosen adjusting screw 1 revolution.
3. 10 N•m [ 90 in-lb ]



Hold the adjusting screw in position. The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the lock nut.

### Torque Value:

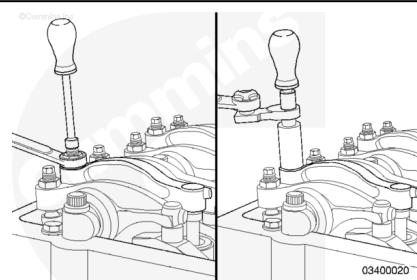
With Torque Wrench Adapter, Part Number ST-669

1. 45 N•m [ 33 ft-lb ]

### Torque Value:

Without Adapter

1. 60 N•m [ 44 ft-lb ]

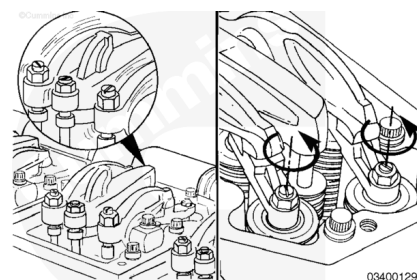


## Crossheads

Crosshead adjustment **must always** be completed before attempting to adjust the valves.

Adjust the crossheads on the cylinder that has both valves closed.

Loosen the crosshead adjusting screw lock nuts on the intake and exhaust valve crossheads.

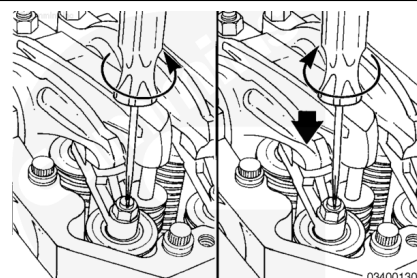


Use this procedure to adjust both the intake and the exhaust crossheads.

Turn the adjusting screw out a minimum of one turn.

Hold the crosshead down against its mating valve stem.

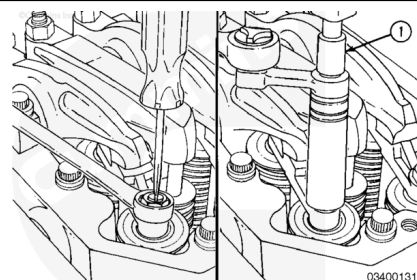
Turn the adjusting screw in until it touches the top of the valve stem but does **not** raise the crosshead.



Hold the adjusting screw in position. The adjusting screw **must not** turn when the lock nut is tightened to its torque value.

Tighten the lock nut.

Torque values are given when using and **not** using adapter, overhead adjustment kit, Part Number 5298979 (1).



#### Torque Value:

Without Jacobs® Brake (With Adapter)

1. 35 n•m [ 25 ft-lb ]

#### Torque Value:

Without Jacobs® Brake (Without Adapter)

1. 40 n•m [ 30 ft-lb ]

#### Torque Value:

With Jacobs® Brake (With Adapter)

1. 40 n•m [ 30 ft-lb ]

#### Torque Value:

With Jacobs® Brake (Without Adapter)

1. 45 n•m [ 33 ft-lb ]

## Valves

Use the table below to for valve adjustment initial set.

| Valve Adjustment (Initial Set) |                            |
|--------------------------------|----------------------------|
| Reference Point                | Valve                      |
| A                              | Exhaust 0.69 mm [0.027 in] |

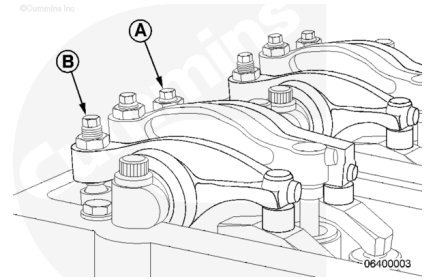
### Valve Adjustment (Initial Set)

| Reference Point | Valve                     |
|-----------------|---------------------------|
| B               | Intake 0.36 mm [0.014 in] |

The recheck limits are to be used when checking the overhead set during troubleshooting or periodic maintenance. The initial set specifications **must** be used when the valves are being adjusted.

### Valve Recheck Limits

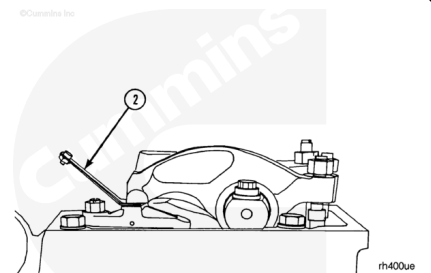
| Reference Point | Valve                                  |
|-----------------|----------------------------------------|
| A               | Exhaust minimum of 0.610 mm [0.024 in] |
| A               | Exhaust maximum of 0.762 mm [0.030 in] |
| B               | Intake minimum 0.279 mm [0.011 in]     |
| B               | Intake maximum 0.432 mm [0.017 in]     |



Make sure the crosshead is firmly in place on the valve stem tips.

Select a feeler gauge for the correct valve lash specification. Use service tool, Part Number 3163171 (intake) or Part Number 3163172 (exhaust).

Insert the feeler gauge between the rocker lever socket and crosshead.



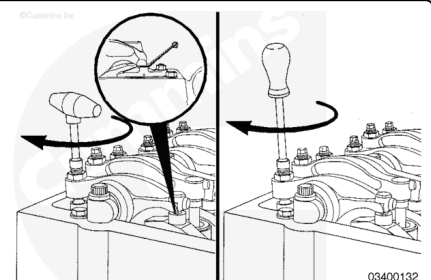
There are two different methods for establishing valve lash clearance. Either method can be used. The torque wrench method has proven to be the most consistent.

- Torque wrench method: Tightens the adjusting screw to 0.68 N•m [6 in-lb] against the feeler gauge, using torque wrench, Part Number 3376592.
- Feel method: Turns the adjusting screw until the lever touches the feeler gauge using a nut driver.

This step outlines the procedure for setting the valve lash using the torque wrench method.

Make sure the parts are in alignment and squeeze the oil out of the valve and injector train while tightening the adjusting screw.

Loosen the adjusting screw one revolution.



Insert the appropriate feeler gauge between the rocker lever and the crosshead.

Tighten the adjusting screw.

**Torque Value:** 0.68 n•m [ 6 in-lb ]

Remove the feeler gauge.

The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the lock nut.

**Torque Value:**

With Torque Wrench Adapter, Part Number ST-669

1.45 n•m [ 33 ft-lb ]

**Torque Value:**

Without Adapter

1.60 n•m [ 44 ft-lb ]

This step outlines the procedure for setting the valve using the feel method.

Make sure the parts are in alignment and squeeze the oil out of the valve and injector train while tightening the adjusting screw.

Loosen the adjusting screw one revolution.

Insert the appropriate feeler gauge between the rocker lever and crosshead.

The feeler gauge **must** slide backward and forward with **only** a slight drag.

The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the lock nut.

**Torque Value:**

With Torque Wrench Adapter, Part Number ST-669

1.45 n•m [ 33 ft-lb ]

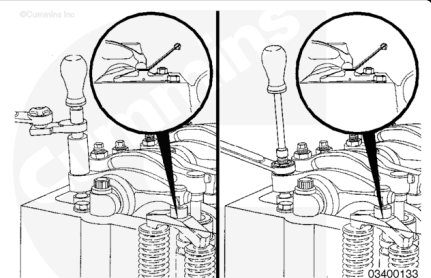
**Torque Value:**

Without Adapter

1.60 n•m [ 44 ft-lb ]

Attempt to insert a feeler gauge 0.03 mm [0.001 in] thicker than used for setting the valve lash.

The valve lash is **not** correct if the feeler gauge will fit.



Repeat the adjustment process until the clearance is correct on both the intake and exhaust valves.

## Finishing Steps

- Install the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/18/18-003-011-tr.html).



## Worksheet

- The engine firing order is 1-5-3-6-2-4.
- The cylinders are numbered from the front gear cover end of the engine.
- Two crankshaft revolutions are required to adjust all of the valves and the injectors.
- One pair of valves and one injector are adjusted at each pulley index mark before rotating the engine to the next index mark.
- The valves and the injectors on the same cylinder are **not** adjusted at the same mark.
- Each cylinder has three rocker levers. The rocker lever nearest the front of the engine is the exhaust rocker lever.

The table below is provided for recording values.

### Inspection Report for Valve Lash and Injector Torque

| Work Sequence | Valves to be Set | Intake | Exhaust | Reset (Y/N) | Crosshead (OK/Reset) | Cylinder Number | Injector Torque | Reset (Y/N) |
|---------------|------------------|--------|---------|-------------|----------------------|-----------------|-----------------|-------------|
|               | A-5              |        |         |             |                      | 4               |                 |             |
|               | B-3              |        |         |             |                      | 1               |                 |             |
|               | C-6              |        |         |             |                      | 5               |                 |             |
|               | A-2              |        |         |             |                      | 3               |                 |             |
|               | B-4              |        |         |             |                      | 6               |                 |             |
|               | C-1              |        |         |             |                      | 2               |                 |             |

Customer:

Location:

Unit Number: Hours: Date:

Job Number:

Engine Model Number: Engine Serial Number:

Type of Inspection (Failure, Routine, Complaint):

Inspection Notes:

**Note : This page can be copied for convenience.**